

OTOPAIR

INTERNAL TOOL SPECIFICATION · DOCUMENT 1 OF 3

Ops Portal

The B2C command center. Every user, every booking, every dollar, every conversation — one workspace where anyone on the team can walk in,

navigate, and understand the full customer lifecycle without asking an engineer.

VERSION	v1.0 — Build Specification	DATE	July 11, 2026
BUILD OWNER	Daniel C. (shop portal / admin)	DESIGN	AB
BACKEND	Convex (production deployment) · Clerk auth · Next.js at otopair.com/shop/admin	STATUS	Ready for sprint scoping

1 Purpose & Philosophy

The Ops Portal is where Otopair runs the **consumer side of the marketplace**: users, their garages, their bookings, their money, their conversations with Oto, and their rewards. Today this lives as a handful of surface-level directory pages inside one all-in-one admin. The problem isn't missing pages — it's missing **depth and connectedness**. An operator looking at a booking cannot see the payment's error history, the AI conversation that created it, or the follow-up it should have generated, without four separate lookups.

The design principle carries over from the consumer app: **"If a screen feels like selling, it's wrong. If it feels like a dashboard protecting an asset, it's right."** Internally, the asset being protected is *operational truth* — every screen must answer "what actually happened, and what happens next?" in one place. Three rules govern every page in this document:

- **Everything is a link.** Any entity name rendered anywhere (user, booking, shop, VIN, payment) is a clickable chip that deep-links to its detail page — including across portals.
- **Every state change has a story.** Status is never shown as a bare label; it always expands into its append-only history ([booking_status_history](#), [payment_status_history](#)) with actor and reason.
- **Reads are free, writes are ceremonies.** Any admin mutation requires a reason field and lands in the audit log (locked team decision, Apr 20). No silent edits, ever.

2 Users & Permissions

2.1 Personas

PERSONA	TEAM MEMBER(S)	WHAT THEY COME HERE TO DO
Super Admin	Temur, AB	Everything. Full read/write across all three portals, role management, destructive actions (with reason).
Ops Admin	Rotating (launch phase: Temur + Daniel)	Daily booking triage, payment issue resolution, user support lookups, review moderation, follow-up management.
Support (read + limited write)	Front-desk / future support hires	Look up a user or booking during a phone call; annotate; trigger a follow-up. Cannot touch money or delete anything.
Read-only	Ahmad, Bilal, Yassin (when debugging consumer flows / pulling metrics)	See everything in Ops scope, change nothing. Bilal's marketing metrics live in Analytics pages.

2.2 Permission matrix (Ops scope)

CAPABILITY	SUPER ADMIN	OPS ADMIN	SUPPORT	READ-ONLY
View users / bookings / payments / analytics	✓	✓	✓	✓
Edit user profile fields, resend verification	✓	✓	✓	—
Change booking status (cancel / reschedule assist)	✓	✓	—	—
View payment detail + Stripe link-out	✓	✓	✓ (masked amounts optional)	✓
Adjust reward wallet balance (credit grants)	✓	✓ (cap \$50)	—	—
Process account deletion queue	✓	✓	—	—
Hide / restore a review	✓	✓	—	—
View audit log	✓	✓	own actions only	—

Implementation

Roles live in Clerk as organization roles (`super_admin`, `ops_admin`, `support`, `readonly`, `data_admin`, `shop_success`), enforced twice: Next.js middleware gates routes; every Convex admin mutation re-checks the role server-side. UI hides what middleware would block — no dead buttons.

3 Information Architecture

One left sidebar, seven groups, sixteen destinations. The portal switcher (Ops / Shops / Data) sits at the very top of the sidebar as a segmented control — switching portals swaps the entire nav tree but preserves %K and the audit trail.

```
OPS — portal switcher: [ Ops | Shops | Data ]
├─ Overview /ops
├─ PEOPLE
│  ├─ Users /ops/users
│  │  └─ User Detail /ops/users/:id (9 tabs)
│  └─ Deletion Queue /ops/users/deletions
├─ MARKETPLACE
│  ├─ Bookings /ops/bookings (board + list)
│  │  └─ Booking Detail /ops/bookings/:id
│  └─ Reviews /ops/reviews
├─ MONEY
│  ├─ Payments /ops/payments
│  │  └─ Payment Detail /ops/payments/:id
│  ├─ Transactions Ledger /ops/transactions
│  └─ Rewards /ops/rewards (wallets · ledger · deals · claims · tiers)
├─ ENGAGEMENT
│  ├─ Oto AI /ops/oto (conversations · transcripts)
│  └─ Follow-ups /ops/follow-ups
├─ INSIGHT
│  ├─ Analytics /ops/analytics (events · funnels)
│  └─ System Health /ops/health (client_logs)
└─ GOVERNANCE
   └─ Audit Log /ops/audit
```

3A Screen Anatomy — the visual system

Every screen in all three portals is assembled from one shell and five templates. This section defines them *visually* so a page's look is never invented twice. Tokens are the locked brand set; measurements are the build values.

The shell — always present

ELEMENT	VISUAL SPECIFICATION
Sidebar	240px fixed, gradient #0B1B3F → #0E2148. Top: portal switcher — segmented control 3 cells (Ops / Shops / Data), active cell #2563EB white text, radius 10px, 13px SF Pro Semibold. Groups: 11px uppercase #5E76A8, letter-spacing .08em, 20px top margin. Items: 14px SF Pro Medium #C7D6F9, 36px row height, 12px left pad, lucide icon 16px; active item = #2563EB pill (radius 10px) white text; hover = rgba(37,99,235,.25). Bottom-pinned: admin avatar + role label 12px.
Top bar	56px, white, bottom border #E2E8F0. Left: breadcrumb 13px #6B7280 (portal / section / entity). Right: 🔍 search pill (#F8FAFC, border #E2E8F0, radius 10px, "Search... 🔍" 13px #9CA3AF), notification bell, avatar 28px.
Env banner	Non-production only: 32px amber #FEF3C7 strip above top bar, text 12px #B45309 "⚠ {deployment} — not production". Production renders nothing.
Content area	Background #F8FAFC, 24px padding, max-width 1360px centered. Cards: white, radius 16px, border #E2E8F0, shadow 0 1px 2px rgba(16,24,40,.05), 20px padding.
Page header	Title 22px SF Pro Semibold #1A1A1A + live-count pill (#EFF6FF bg, #1D4ED8 text, 12px, radius 999). Right-aligned actions: primary button #2563EB white 14px radius 12px; secondary ghost white/border #E2E8F0 #374151.
Filter row	Chips 13px radius 10px: idle white/border #E2E8F0 #374151; active #EFF6FF/#1D4ED8 with ×. Right end: saved-views dropdown + column picker icon.
Data table	Header row 12px uppercase #6B7280 on #F8FAFC, 40px; body rows 44px 14px #1A1A1A, zebra #F8FAFC on even, hover #EFF6FF@40%; sortable headers get ↑↓ 10px; status pills 11px SF Semibold radius 999 (colors per state); row click opens right drawer, never navigates away for quick looks.
Right drawer	480px slide-over, white, left border 2px #2563EB, 24px padding; header = entity chip + status + close ×; sticky footer for actions. Full detail pages exist for deep work; the drawer is the 10-second look.
Write ceremony modal	520px centered, radius 20px: change summary as before → after diff table; reason textarea (min 10 chars, counter); Confirm disabled until valid, danger actions get red primary #B91C1C.
States	Loading = skeleton bars #E2E8F0 shimmer; Empty = 48px lucide icon #9CA3AF + one line 14px #6B7280 + primary action; Error = retry + copy-error-id; Partial = amber "orphaned" chip.

Ops sidebar — exact tabs, in order

Ops | Shops | Data

Overview

PEOPLE

Users

Deletion Queue

MARKETPLACE

Bookings

Reviews

MONEY

Payments

Transactions

Rewards

ENGAGEMENT

Oto AI

Follow-ups

INSIGHT

Analytics

System Health

GOVERNANCE

Audit Log

Ops / Overview

Search... %K

KPI ROW — T1

Bookings today

GMV today

Platform rev

Active 7d

Failed pay 24h

Pending del.

LIVE ACTIVITY FEED

NEEDS ATTENTION

FUNNEL SNAPSHOT

started

service

shop

time

payment

done

The Ops shell with T1 (Dashboard) in the content area — this is /ops exactly as built.

The five templates

T2 — LIST

Page Title 1,204

+ Action

filter

filter

filter

date ▼

saved ▼

< 1 2 3 ... > 50/page

T2 + RIGHT DRAWER (ROW CLICK)

Entity chip status

Primary action

T3 — DETAIL WITH TABS

Entity Name status chips · chips · chips

Tab 1 Tab 2 Tab 3 Tab 4 ...

T4 — BOARD (KANBAN)

COL A · 4

COL B · 2

COL C · 6

COL D · 1

T5 — WORKSPACE (THREE-PANE; REVIEW QUEUE, TRANSCRIPTS)

QUEUE

↑ ↓ navigate

EVIDENCE / CONTENT

A pprove

R eject

E dit

CONTEXT

3B Page Layout Atlas — every Ops page, top to bottom

Nav position → template → zones in render order. "You see" lines are build-exact.

Overview T1 · sidebar item 1 · /ops

KPI row	6 equal cards: value 28px SF Semibold #1A1A1A, label 12px #6B7280, delta arrow 12px (green #059669 / red #B91C1C). Failed-payments & pending-deletions cards turn amber border when >0.
Two-column	Left 58%: Live activity feed — timestamp 12px #9CA3AF, event line 14px with inline chips, new rows fade in 300ms. Right 42%: Needs Attention — amber-bordered card, each row = icon + one-liner + "Fix →" link to the surface.
Funnel strip	6 stage blocks with % drop between; hover shows median time-in-stage.

Users T2 · PEOPLE → Users · /ops/users

Header	"Users" + count pill · right: Export CSV (ghost).
Filters	Onboarding ▼ · Verified ▼ · Has vehicle · Smartcar · Provider ▼ · Created ▼ · △ Pending deletion toggle.
Table	Avatar+Name · Email ✓ · Phone ✓ · Vehicles # · Bookings # · Last active · Onboarding pill · Flags. Row click → drawer (identity card, 3 quick stats, "Open full profile").

User Detail T3 · /ops/users/:id · 9 tabs

Header card	64px avatar · name 22px · username 13px #6B7280 · verification badges · Clerk ID mono 12px with copy · right: "Open in Clerk" ghost + overflow ; (Edit, Follow-up).
Tab strip	Profile · Garage · Bookings · Money · Rewards · Oto AI · Onboarding · Activity · Danger Zone (red text, always last).
Garage tab	One card per vehicle: 96px vehicle image left; nickname 16px + VIN mono chip; spec line 13px (year make model trim engine); mileage + primary ★; Smartcar dot (green/grey/red); below: odometer sparkline 120×32, collapsible Health Snapshots + Maintenance Records lists.
Money tab	Two stacked cards: Payments table (5 col) then Transactions ledger styled exactly like the consumer app rows (icon, description, signed amount green/red) — support sees what the user sees.
Danger Zone	Red-tinted card #FEF2F2 border #FECACA: deletion request state + survey answer; notification prefs toggles (read-only); "Process deletion" red button → checklist modal (4 steps, each its own ceremony).

Deletion Queue T2 · PEOPLE → Deletion Queue · /ops/users/deletions

Header	"Deletion Queue" + count pill + permanent right-aligned stat "Oldest request: N days" (turns red >25d).
Table	User · Requested · Age badge (amber >7d, red >25d) · Survey response (or "skipped" grey) · open-bookings blocker column. Drawer = the 4-step processing checklist with per-step Complete buttons.

Bookings T4 + T2 · MARKETPLACE → Bookings · /ops/bookings

View toggle	Segmented Board List, top-right; Board default 7am–9pm.
Board	Columns: Confirmed · In Progress (cards carry live_stage sub-badge: confirmed / in service / vehicle ready) · Completed today · Cancelled today. Card 260px: time 13px bold · user chip · vehicle chip · shop+mechanic line · services summary · total right-aligned 14px semibold. No drag — the board is a window, not a lever.
List	13 columns per \$5.4; payment dot column far right (green/amber/red/—).

Booking Detail T3 · /ops/bookings/:id

Header	Status pill + live_stage · scheduled date/time 16px · chips row: user / vehicle / shop / mechanic / time-slot.
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Left column 60%	① Status timeline — vertical line, node per transition, actor chip + reason 13px, timestamps right. ② Services & Money — line items, then cost stack right-aligned; platform fee its own row; total 18px semibold; red "INTERNAL ONLY" watermark tag on the card.
Right column 40%	③ Estimate vs Actual (when job_actuals): two-column est/actual with variance % badge; parts_used chips; amber variance banner → Data portal. ④ Payment card. ⑤ Around this booking: review, follow-ups, origin Oto chat, funnel record — each one row + chevron.
Footer bar	Sticky: Cancel booking (ghost-red, ceremony w/ reason enum) · Add note · Trigger follow-up.

Payments T2 + Payment Detail T3 · MONEY → Payments

List	Default filter = Failed. Columns incl. Stripe PI mono with copy icon; duplicate-idempotency rows render red left border (incident signal).
Detail	Left: field card + status timeline with error_code/message rows in red mono 12px. Right: linked booking + transaction cards. Header action: "Open in Stripe" primary. Zero mutation buttons anywhere — forensic window by design.

Transactions T2 · MONEY → Transactions · /ops/transactions

Header	Title + "Reconciliation mode" toggle switch right.
Table	Ledger rows, signed amounts. Reconciliation ON: mismatch rows highlight amber with reason chip (amount≠payment / orphan transaction / orphan payment); summary bar on top "N mismatches" green when zero.

Reviews T2 · MARKETPLACE → Reviews · /ops/reviews

Default view	"Needs eyes" = rating ≤3, newest first; star rendering 16px #F59E0B.
Row	Stars · comment 2-line clamp (click expands in place) · user/shop/mechanic/booking chips · Hide (ghost-red, ceremony). Hidden rows collapse to grey strip "Hidden — reason — Restore".

Follow-ups T2×3 views · ENGAGEMENT → Follow-ups

Metrics strip	Pending · Sent 7d · Sent → booking conversion % (3 inline stat cards above filters).
Views	Queue (pending by scheduled_for, next-to-fire on top, countdown chip) · Sent · Dismissed. Drawer: full message in phone-frame preview, Edit (pending only), Send now, Cancel.
Create	Right-side wizard: user → vehicle → service → type → schedule → message with template suggestions; live phone preview updates as you type.

Oto AI T2 → T5 · ENGAGEMENT → Oto AI · /ops/oto

KPI strip	Conversations 7d · avg messages · → booking % · top scenario chips.
Transcript viewer	T5 variant: left = conversation list; center = chat bubbles (user right #EFF6FF, Oto left white, system collapsed grey); right = per-message metadata (confidence, intent, suggestion chips). PII banner 12px amber pinned top: "Private user conversation — access logged." Read-only, no composer.

Rewards T3 (5 sub-tabs) · MONEY → Rewards · /ops/rewards

Tabs	Wallets · Credit Ledger · Deals · Claims · Tiers.
Ledger	Type chips color-coded (earn green family / redeem blue family / earn_manual purple "ADMIN"); expiry countdown column turns amber ≤30d; saved view "Expiring ≤30d" pinned.
Deals	Left: editable table. Right: live phone-frame rendering the deal card exactly as in-app; Publish = ceremony.

Analytics · System Health · Audit T2 variants · INSIGHT / GOVERNANCE groups	
Analytics	Events: filter row + hourly volume bar chart (120px) above table; row expands JSON viewer. Funnels: stage bars with red drop-off labels + week-over-week ghost bars; drop_off_reason donut right.
Health	Error-rate sparkline header; grouped-by-signature list, count badge, expand → stack mono 11px scrollable; "View user's session" link per row.
Audit	When · Actor chip · Portal badge · Action verb · Entity chip · Reason (italic) · diff expander (green/red rows). Immutable; export ghost button.

4 Portal Shell Standards (shared by all three portals — full spec here, referenced by Docs 2 & 3)

STANDARD	SPECIFICATION
⌘K Global Search	One command palette across all portals. Typing searches: users (name/email/phone via <code>by_clerkUserId</code> + client filter), VINs (<code>vehicles.by_vin</code>), bookings (id), shops (name), mechanics, OEM part numbers (<code>oem_parts.by_part_number</code>). Results grouped by entity type with portal badge; Enter deep-links. Recent entities pinned per admin.
Entity Chips	UserChip, BookingChip, ShopChip, MechanicChip, VinChip, PartChip, PaymentChip. Each renders: icon + primary label + status dot + hover card (3–5 key fields) + click → detail route. Chips are the connective tissue between portals: a VinChip inside a booking (Ops) opens the vehicle in the Data portal. If the viewer's role lacks that portal, chip renders with lock tooltip instead of dead link.
Tables	Server-paginated via Convex <code>usePaginatedQuery</code> on named indexes only — never full scans. Default 50 rows. Column sort limited to indexed fields (greyed otherwise, with tooltip "not indexed"). Column picker + saved views per admin (<code>saved_views</code> , proposed). Every table exports CSV of the current filter (read-only export; audit-logged).
Realtime	Convex reactivity means every list and dashboard is live — no refresh button anywhere in the product. New rows glide in with a 300ms fade; counters tick. This is the "wow" of the internal tool and costs nothing extra.
Status History Pattern	Any status pill anywhere expands (click) into a right-side drawer with the append-only history: old → new, actor chip, reason, timestamp. Powered by <code>booking_status_history</code> / <code>payment_status_history</code> and <code>admin_audit_log</code> for admin-side edits.
Write Ceremony	Every mutation modal has: summary of the change (before → after diff), a required reason field (min 10 chars), confirm button disabled until reason present. Mutation writes the domain change + one <code>admin_audit_log</code> row atomically in the same Convex mutation.
States	Every page defines all four: loading (skeleton rows, brand-blue shimmer), empty (illustrated, one-line explanation + primary next action), error (retry + copy-error-id), partial (data present but linked entity missing — render chip in amber "orphaned" state, never crash).
Design tokens	Primary Blue #2563EB · Blue Dark #1D4ED8 · Blue Light #EFF6FF · Success #059669 / #ECFDF5 · Text #1A1A1A / #374151 / #6B7280 / #9CA3AF · Border #E2E8F0 · Surface #F8FAFC · Card radius 16px · Button radius 12px · SF Pro. Density: internal tables run 8px tighter vertical rhythm than consumer app.
Environment banner	Non-production deployments (ahmad, daniel, waleed, temurbek) render a persistent amber top bar with deployment name. Production renders nothing. Prevents the classic "edited the wrong environment" incident.

5 Page-by-Page Specification

5.1 Overview — /ops

Purpose: 30-second answer to "is the marketplace healthy right now?" Optimized for the morning glance and the launch-phase obsessive refresh (which, being Convex-reactive, is automatic).

ZONE	CONTENTS	DATA SOURCE
KPI row (6 cards)	Bookings today · GMV today (Σ total_cost) · Platform revenue today (Σ platform-fee component) · Active users 7d · Failed payments 24h · Pending deletions	bookings by_created_at / by_scheduled_date, payments by_status, users by_isPendingDeletion, analytics_events
Live activity feed	Reverse-chron stream of: booking created/status change, payment completed/failed, review posted, deletion requested. Each row = timestamp + entity chips + one-line event.	booking_status_history, payment_status_history, reviews merged client-side by changed_at
Needs attention	Auto-populated task list: payments failed >1h, bookings in_progress past scheduled end +2h, follow-ups pending & overdue, deletion requests >7 days old. Each row links to the fix surface.	payments, bookings, follow_ups by_status_and_scheduled, users
Funnel snapshot	Booking-flow funnel last 7d: started → service_selected → shop_selected → time_selected → payment_info → completed, with % drop per stage.	conversion_funnels by_funnel_type + by_stage

Empty state (launch week reality): KPI cards render "0" honestly — no fake sparkle. Feed shows "Quiet so far today." Trust-first applies internally too.

5.2 Users — /ops/users

Purpose: Find any user in <5 seconds; understand their whole relationship with Otopair in one detail page.

List columns: Avatar+Name (UserChip) · Email (+✓ if emailConfirmed) · Phone (+✓ if phoneVerified) · Vehicles (count from vehicle_owners by_user_status active) · Bookings (lifetime) · Last active (lastUpdated) · Onboarding (Completed / In progress / Tell-us pending) · Flags (△ pending deletion).

Filters: onboarding state · verification state · has active vehicle · has Smartcar connection · signup provider (auth_provider) · created date range · pending deletion. **Search:** name / email / phone / username (client-side over paginated window + ⌘K for exact).

Row actions: Open · Copy email · Start follow-up · (Ops Admin+) Edit profile.

User Detail — /ops/users/:id — nine tabs

TAB	CONTENTS & BEHAVIOR	TABLES
Profile	Identity card (photo, name, username, email/phone + verification badges, auth provider, Clerk ID with copy, language/units, created/updated). Inline edit for name/phone/language with write ceremony. "Open in Clerk" link-out.	users

Garage	One card per <code>vehicle_owners</code> row (active + removed shown separately): vehicle image (<code>image_url</code>), nickname, VIN (VinChip → Data portal), year/trim/engine resolved through <code>vehicles-trims-models-makes</code> , mileage, primary star, driving profile (<code>avgMonthlyDriving</code> / <code>drivingConditions</code> / <code>knownIssues</code>), Smartcar status (<code>connectionStatus</code> , <code>connectedAt</code> ; token health from <code>smartcar_connections</code>), onboarding phase. Sub-panels per vehicle: odometer sparkline (<code>odometer_history</code> by <code>vehicle_and_date</code>), health snapshots timeline (<code>vehicle_health_snapshots</code>), user-reported maintenance (<code>maintenance_records</code> by <code>vehicle_and_type</code>).	<code>vehicle_owners</code> , <code>vehicles</code> , <code>smartcar_connections</code> , <code>odometer_history</code> , <code>vehicle_health_snapshots</code> , <code>maintenance_records</code>
Bookings	Compact booking list scoped via <code>bookings.by_user_id</code> — status pill (expands to history), shop chip, services, date, total. Row → Booking Detail.	<code>bookings</code>
Money	Two stacked panels: Payments (<code>payments.by_user_id</code>) and user-facing Transactions ledger (<code>transactions.by_user_id_created_at</code> desc) — exactly what the user sees in-app, so support can say "I see the same screen you do."	<code>payments</code> , <code>transactions</code>
Rewards	Wallet balance + auto-apply toggle state + miles_safe (<code>user_reward_wallets</code>), credit ledger (<code>ownership_credit_transactions</code> by <code>user_id_created_at</code>) with expiry countdown per earn row, contribution claims (<code>user_contribution_claims</code>), per-vehicle tier (<code>vehicle_tiers</code>). Ops Admin: "Grant credit" action (amount ≤ cap, reason required, writes ledger row typed <code>earn_manual</code> — enum addition, see §7).	<code>user_reward_wallets</code> , <code>ownership_credit_transactions</code> , <code>user_contribution_claims</code> , <code>vehicle_tiers</code>
Oto AI	Conversation list (<code>ai_conversations.by_user_id</code>): started, duration, message_count, scenario_detected, led_to_booking ✓ (BookingChip when linked). Row → transcript viewer (§5.9).	<code>ai_conversations</code>
Onboarding	The intelligence goldmine, currently invisible to the team: full Q&A array, user_intentions, car_knowledge_level (1–3), from <code>onboarding_questions_answers.by_user_id</code> . Read-only.	<code>onboarding_questions_answers</code>
Activity	Event stream (<code>analytics_events.by_user_id</code>) + this user's funnels (<code>conversion_funnels.by_user_id</code>) + their client-side errors (<code>client_logs.by_user_id</code>) — support can see the crash the user is describing.	<code>analytics_events</code> , <code>conversion_funnels</code> , <code>client_logs</code>
Danger Zone	Deletion request state (requestedAt, survey response/skipped), notification prefs (<code>user_settings_preferences</code>), and the Process Deletion action (Ops Admin+, double-confirm + reason). Red-tinted panel; everything here is audit-logged.	<code>users</code> , <code>user_settings_preferences</code>

5.3 Deletion Queue — </ops/users/deletions>

Purpose: Compliance surface. Every user with `isPendingDeletion = true` (`users.by_isPendingDeletion`), sorted by `deletionRequestedAt` asc, with age badge (amber >7d, red >25d). Columns: UserChip · requested date · age · survey response (or "skipped") · open bookings blocker (if any booking not in terminal status — deletion blocked until resolved). Detail drawer shows the full checklist the processor must complete: cancel/settle open bookings → confirm no pending payments → anonymize PII → mark processed. Each step is a discrete audited action. **SLA note:** the page header shows "oldest request: N days" permanently — the queue polices itself.

5.4 Bookings — </ops/bookings>

Purpose: The heartbeat page. Two views, toggle top-right; board is default during business hours.

Board view: Kanban columns by status — Confirmed → In Progress (cards show `live_stage` sub-badge: `booking_confirmed` / `service_in_progress` / `vehicle_ready`) → Completed (today) → Cancelled (today). Cards: time, UserChip, VinChip, ShopChip + mechanic, services summary, total. Card click → detail. *No drag-to-change-status* — status flows from the shop portal / app; Ops intervenes only through the explicit actions in detail view. The board is a window, not a lever (trust the source of truth).

List view columns: ID · Created · Scheduled (date+time) · UserChip · VinChip · ShopChip · Mechanic · Services (count, hover list) · Est. minutes · Labor \$ · Parts \$ · Total \$ · Status (pill → history drawer) · Payment (dot: paid/pending/failed/none).

Filters: status (`by_status`) · shop (`by_shop_id` / `by_shop_and_status`) · date range (`by_scheduled_date`) · user (`by_user_id`) · has payment issue.

Booking Detail — </ops/bookings/:id>

PANEL	CONTENTS
Header	Status pill + <code>live_stage</code> , scheduled date/time, created/updated, UserChip · VinChip · ShopChip · MechanicChip, time-slot chip (date, start–end from <code>time_slots</code>).
Timeline	Vertical status timeline from <code>booking_status_history</code> <code>by_booking_id</code> : every transition with actor (system vs user vs admin) and reason. The single most useful support artifact in the portal.
Services & Money	Line items: each service (name via <code>services</code> , options if any), then cost stack — <code>labor_cost</code> , <code>parts_cost</code> , taxes & fees, platform fee (7%) as its own line , <code>total_cost</code> . INTERNAL ONLY This breakdown never leaves the portal; the pricing construction is not customer-facing beyond the fee line the app already shows.
Estimate vs Actual	When <code>job_actuals</code> <code>by_booking_id</code> exists: side-by-side <code>estimated_labor_minutes</code> vs <code>actual_labor_minutes</code> , <code>parts_est</code> vs <code>actual_parts_cost</code> , <code>parts_used</code> list (name, OEM# → PartChip, cost), difficulty (1–5), <code>technician_notes</code> , started/completed timestamps. Variance % badge; if the linked <code>spec_variances</code> row is flagged, amber banner links to Data portal variance page — this is where Ops meets the data flywheel.
Payment	PaymentChip → payment detail; inline status, method, <code>stripe_payment_intent_id</code> (copy + "Open in Stripe"), error history summary if any transitions failed.
Around this booking	Review (if posted), follow-ups generated (<code>follow_ups.by_booking_id</code>), origin AI conversation (<code>ai_conversations.by_booking_id</code> — "This booking started in an Oto chat" with transcript link), funnel record (<code>conversion_funnels.by_booking_id</code>), spec confirmation from user if given.
Actions (Ops Admin+)	Cancel booking (reason enum: <code>user_requested</code> / <code>shop_requested</code> / <code>no_show</code> / <code>other</code> + free text; frees the time slot; writes status history with <code>changed_by</code>) · Add internal note (proposed <code>admin_notes</code> , §7) · Trigger follow-up. <i>Refunds are intentionally absent — Stripe-hosted flows own refunds/disputes (locked decision, Apr 23); the portal deep-links out.</i>

5.5 Payments — </ops/payments> & Payment Detail

List: Created · PaymentChip (id) · BookingChip · UserChip · ShopChip · Amount · Method (card/apple_pay/...) · Status pill · Stripe PI (mono, copy). Filters: status ([by_status](#) — **failed is the default landing filter**), method, date ([by_created_at](#)), shop, user. Duplicate-guard indicator when two rows share idempotency_key ([by_idempotency_key](#)) — should never happen; if it renders, it's an incident.

Detail: full field card · status timeline from [payment_status_history](#) [by_payment_id](#) including error_code / error_message per failed hop (insufficient_funds, card_declined...) · linked booking panel · linked [transactions](#) row ([by_payment_id](#)) confirming what the user's app displays · "Open in Stripe" primary action. **No mutation actions here at all** — money state changes only via Stripe webhooks; the portal is a forensic window. This is a feature, not a gap: one-way doors stay one-way.

5.6 Transactions Ledger — </ops/transactions>

The user-facing money feed, admin-wide. Columns: Created · UserChip · Description / sub_description · Type (charge/credit/refund) · Amount (signed, red/green) · Status · linked Booking/Payment/Shop chips · icon_type. Filters: type ([by_user_id_type](#) when user-scoped; global list uses date pagination), status, date. **Reconciliation mode** (toggle): joins each charge transaction to its [payments](#) row and flags mismatches (amount ≠ payment.amount, transaction without payment, payment without transaction) — a daily 2-minute integrity sweep, one glance.

5.7 Reviews — </ops/reviews>

Columns: Created · Rating (stars, filter via [by_rating](#)) · Comment (2-line clamp, click expands) · UserChip · ShopChip · MechanicChip · BookingChip. Filters: rating ≤3 (default view: "needs eyes"), shop, mechanic, date. Actions (Ops Admin+): Hide review / Restore (soft flag — proposed is_hidden field or moderation table, §7; hiding requires reason, audit-logged; hidden reviews excluded from shop/mechanic aggregates by the read path). **Anti-pattern guard:** there is deliberately no "edit review" and no "boost rating" anywhere. We moderate abuse; we never manufacture trust.

5.8 Follow-ups — </ops/follow-ups>

Purpose: The retention engine's control room. Views: **Queue** (pending, sorted by scheduled_for via [by_status_and_scheduled](#) — "what fires next"), **Sent, Dismissed/Completed**. Columns: Scheduled for · Type (reminder / maintenance_due / inspection) · UserChip · VinChip · Service · Message preview · Status · Origin BookingChip. Detail drawer: full message, timestamps (created/sent), edit-before-send (pending only, ceremony), Cancel, Send-now override. **Create follow-up** action (also reachable from User/Booking detail): pick user → vehicle → service → type → schedule → message (template suggestions per type). Metrics strip on top: pending count · sent 7d · sent → booking conversion (join follow_ups to subsequent bookings for same VIN+service within 30d — computed query).

5.9 Oto AI — </ops/oto>

Conversations list: Started ([by_started_at](#)) · UserChip · Duration · Messages (count) · Scenario detected (price_check / booking_flow / ...) · Led to booking ✓ (BookingChip). Filters: scenario, led_to_booking, date. KPI strip: conversations 7d · avg messages · conversion to booking % · top scenarios.

Transcript viewer (</ops/oto/:conversationId>): chat-style rendering of [ai_messages](#) [by_conversation_id](#) — user right, Oto left, system collapsed; per-assistant-message metadata drawer: confidence_score, intent_detected, service/shop suggestions rendered as chips. This is where the team reads real conversations to tune Oto — treat it as the product-improvement reading room. **READ-ONLY** always; PII notice banner reminds viewers these are private user conversations, access is logged.

5.10 Rewards — </ops/rewards>

Five sub-tabs. **Wallets:** UserChip · balance · auto-apply · miles_safe · updated ([user_reward_wallets](#)). **Credit Ledger:** global [ownership_credit_transactions](#) stream — type (earn_service/earn_review/earn_upload/earn_referral/redeem_booking/redeem_giftcard) · amount · UserChip · reference chip · expires_at countdown; filter by type; "expiring ≤30d" saved view for proactive nudges. **Deals:** CRUD on [reward_deals](#) (title, description, credit_amount, price, is_special, service link, display_order) with live phone-frame preview of the deal card; publishing is a ceremony. **Claims:** [user_contribution_claims](#) — double-credit guard view (same user+action+reference twice = red row). **Tiers:** [vehicle_tiers](#) by VIN/user with 12-mo spend; MVP note pinned: "everyone is Driver until V1 activation."

5.11 Analytics — </ops/analytics>

Events explorer: filter by event_type / event_category / date / user / session ([by_event_type](#), [by_event_category](#), [by_session_id](#)); table + hourly volume chart; row expands event_data (booking/shop/service chips, screen_name, custom props JSON). **Funnels:** per funnel_type, stage-by-stage bar with drop-off %, median time-in-stage, drop_off_reason breakdown, date compare (this week vs last). **Session replay-lite:** from any session_id, chronological merge of analytics_events + client_logs for that session — the closest thing to watching the user's screen without a replay tool.

5.12 System Health — </ops/health> · 5.13 Audit Log — </ops/audit>

Health: [client_logs](#) by level ([by_level](#); error default) · message · stack (expand) · UserChip · session · timestamp; error-rate sparkline; "new error signature" grouping by message prefix. Links each error to the user's Activity tab. **Audit:** unified read over [admin_audit_log](#) (proposed §7) + both status-history tables filtered to admin-actor rows: When · Actor (admin chip) · Portal · Action · Entity chip · Reason · before → after diff (expand). Filters: actor, portal, entity type, date. Immutable, exportable, and the first page compliance ever asks for.

6 Cross-Portal Deep Links (Ops edition)

FROM (OPS)	CHIP	LANDS AT
User → Garage → vehicle	VinChip	Data portal · VIN Explorer <code>/data/vins/:vin</code> (specs, enrichment state, provenance)
Booking Detail → shop / mechanic	ShopChip / MechanicChip	Shops portal · <code>/shops/:id</code> · <code>/shops/mechanics/:id</code>
Booking Detail → variance banner	link	Data portal · Verification & Variance <code>/data/verification?job=:jobActualId</code>
Job actuals → parts_used OEM#	PartChip	Data portal · Part Detail <code>/data/parts/:oem</code>

7

Schema Additions Required

PROPOSED — WALEED

TABLE / CHANGE	DEFINITION	WHY
admin_audit_log	actor_clerk_id, actor_name, role, portal, entity_table, entity_id, action, reason, before (any), after (any), created_at · idx: by_created_at, by_actor, by_entity(entity_table, entity_id)	Fulfills the locked Apr 20 audit-log decision as one shared table for all three portals.
admin_notes	entity_table, entity_id, author_clerk_id, body, created_at · idx: by_entity	Internal annotations on users/bookings/shops without polluting domain tables.
saved_views	admin_clerk_id, portal, page, name, filters_json, created_at · idx: by_admin_page	Personal saved filters on every table.
reviews.is_hidden	v.optional(v.boolean()) + hidden_reason, hidden_by, hidden_at	Moderation without deletion; aggregates read-path excludes hidden.
credit type enum	"earn_manual" added to ownership_credit_transactions.type	Admin credit grants must be distinguishable forever.
bookings status enum	Confirm the authoritative status set with Waleed (schema examples show confirmed / in_progress / completed / cancelled; verify whether a pre-payment "pending" exists in code).	The board's columns must match reality, not the doc.

8 Migration From Current Admin & Build Plan

8.1 What already exists → where it goes

CURRENT VIEW (DANIEL, SPRINTS 5–7)	NEW HOME	WORK TYPE
All Users	\$5.2 Users list	Regroup + add columns/filters
Bookings / Booking Detail	\$5.4	Regroup + timeline, estimate-vs-actual, around-this-booking panels
Dashboard home	\$5.1 Overview	Rebuild to KPI + feed + attention model
Audit log (actor+timestamp+reason)	\$5.13 + shared write ceremony	Generalize into admin_audit_log
Role permissions (owner/front desk/mechanic)	\$2 Clerk org roles, extended set	Extend

8.2 Phasing

PHASE	SCOPE	EXIT TEST
P0	Shell (switcher, ⌘K, chips, audit ceremony) · Overview · Users + User Detail (Profile, Garage, Bookings, Money tabs) · Bookings board/list/detail · Payments + detail · Deletion queue	Support can resolve a real customer call end-to-end without touching the Convex dashboard.
P1	Remaining User tabs · Transactions + reconciliation · Reviews moderation · Follow-ups · Oto AI reader	A failed payment, a bad review, and a missed follow-up are each findable and actioned in <2 min.
P2	Rewards suite · Analytics · System Health · saved views · CSV export	Bilal pulls weekly funnel numbers alone; zero engineer time.

9 Definition of Done — 10/10 Checklist

- Every page implements all four states (loading / empty / error / partial) — reviewed against this doc, screenshot per state in the PR.
- Every mutation: reason required, audit row written atomically, visible in `/ops/audit` within one second (reactive).
- Every entity reference anywhere is a chip with hover card and working deep link (or lock tooltip).
- No query without a named index; PR review rejects `.collect()` full scans on unbounded tables.
- Role matrix enforced in middleware *and* mutations; a support-role session literally cannot invoke a money mutation (tested, not assumed).
- Environment banner verified on all five deployments; production shows none.
- Zero dark patterns audit: no fake counts, no manufactured urgency, no vanity metrics — internal tools model the same values as the app.

A Appendix A — Live Production Validation

VALIDATED JUL 11, 2026

Measured read-only against live production and ahmad via Convex MCP. Where this conflicts with the body, the appendix wins.

A.1 The headline finding: production's consumer world is empty

Live production counts: `users` 1 · `bookings` 0 · `payments` 0 · `reviews` 0 · `follow_ups` 0 · `ai_conversations` 0 · `transactions` 0 · `vehicles` 40 · `vehicle_owners` 1. The marketplace demo/test world lives in ahmad (59 users, 77 `vehicle_owners`, 192 vehicles; bookings/payments counts exceed the MCP counting cap). **Consequences:** (1) the empty-state discipline in §4 is not polish — it is the launch-day reality of every Ops page and must be built first-class; (2) TestFlight/partner traffic is not landing in the deployment this portal will read — confirm with Waleed which deployment the live apps point at and whether a production seed/cutover is pending. This is the single most important operational question the validation surfaced.

A.2 Corrections and discoveries

FINDING (MEASURED)	IMPACT
<code>audit_log</code> table already exists (0 rows)	§7 changes from "create <code>admin_audit_log</code> " to extend existing <code>audit_log</code> ; the Apr 20 decision has a head start.
New Ops-relevant tables exist in live schema, absent from the body spec: <code>notification_outbox</code> , <code>app_feedback</code> , <code>bugs</code> , <code>oto_telemetry</code> ; ahmad adds <code>booking_disputes</code> , <code>payment_disputes</code> , <code>booking_approvals</code> , <code>sms_delivery_log</code> , <code>telnyx_*</code> , <code>referrals</code> , <code>user_saved_addresses</code>	Nav additions queued: Notifications outbox joins §5.8's surface; <code>app_feedback</code> + bugs join §5.12 System Health; disputes tables (read-only mirrors — Stripe still owns the flow) join §5.5; SMS delivery log joins Engagement. None change the IA groups.
<code>director_users</code> / <code>director_sessions</code> / <code>director_notes</code> / <code>director_settings</code> exist (0 rows)	Internal-admin auth scaffolding already in schema. §2's Clerk-roles plan must reconcile with it — one decision with Daniel/Waleed on which is canonical, before the shell is built.
Ahmad hot tables return count -1 via MCP	Confirms: all portal counters via indexed Convex queries, never MCP counts (§4 tables standard reaffirmed).

Otopair · Ops Portal Specification v1.0 + Appendix A · Prepared July 11, 2026 · Body verified against canonical schema; Appendix A measured against live deployments July 11, 2026.